

# Cricket Score Prediction

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Machine Learning – Course Project Report

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## 1 Abstract

Cricket is the most popular sport in India, and it is played there in all of its regions in various formats such as T20, ODI and Test. Indian regional teams, the national squad and the international teams all field players in the IPL, a national cricket competition. This league has become well known among cricket enthusiasts due to numerous factors, including live streaming, radio and TV broadcasts. For internet traders and sponsors, the outcome predictions of IPL matches are crucial. As a result, there is a significant market for algorithms that forecast the best score and winning team, which is most crucial.

This project aims to develop an effective machine learning model for predicting the final score of a cricket team in any matches utilizing a real dataset of cricket matches. This project contributes toward improved analytics for cricket matches and offers insights into data-driven forecasting in sports.

## 2 Problem Statement

The aim is to build a machine learning model that predicts the total runs a cricket team will score in a match, given inputs such as toss, current runs, wickets, overs, venue, weather, team's composition, the batting and bowling averages of each player on the team and recent performances. This problem is important for sports analytics, live commentary and strategic decision-making during matches.

## 3 Proposed Methodology

The proposed methodology focuses on predicting the final score of a cricket match using multiple features through regression-based machine learning models. The steps involved in the process are as follows:

- **Data Collection and Preprocessing :**
  - A historical match data is collected from cricket datasets (such as IPL ball-by-ball data)
  - Data is cleaned, categorical variables are label encoded and missing values are handled.
- **Feature Engineering :** In this project, feature engineering played a critical role in improving the prediction accuracy. The raw ball-by-ball data was processed to derive meaningful metrics that capture the game context. Important engineered features include current runs, wickets fallen, overs bowled, runs and wickets in recent overs, and players' batting and bowling averages. Additionally, match conditions such as venue, toss results, and weather were encoded to reflect their influence on scoring patterns. Categorical variables were label encoded, and numerical features were scaled appropriately. Derived features like current run rate, balls remaining, and wickets in hand were computed to better represent the dynamic state of the innings. This comprehensive feature set enabled the models to capture both linear and nonlinear relationships effectively for score prediction.

- **Model Development and Training** : The dataset is split into 70% for training, 20% for validation and 10% for testing. Then five regression techniques were implemented on this dataset :
  - **Linear Regression**  
Linear Regression used to establish a direct linear relationship between feature vector and target vectors.
  - **Ridge Regression**  
An improved model that adds a regularization term to reduce over fitting and it has a tendency to shrink parameters values toward 0.
  - **Lasso Regression**  
Lasso Regression adds an L1 penalty to the OLS loss function and has a tendency to drive some unnecessary coefficients exactly to 0.
  - **Elastic Net Regression**  
Elastic Net combines both L1 and L2 penalties of Lasso and Ridge methods, providing a balance that often leads to superior performance when predictors are highly correlated or when feature selection is desired.
  - **Random Forest Regression**  
Random Forest is an ensemble method using multiple decision trees and averaging their predictions, reducing variance and capturing nonlinear relationships not accessible to simple linear models.
- **Hyperparameter Tuning** : To optimize the performance of each machine learning model, hyperparameter tuning was conducted using grid search and randomized search techniques. For models such as Random Forest, the number of trees, maximum depth, and minimum samples per leaf were varied systematically to find the best combination. In the case of Support Vector Regression, different kernel functions and regularization parameters were tested. This tuning process aimed to balance model complexity and overfitting, ensuring the model could generalize well to unseen match data. Cross-validation was employed during tuning to validate the model on different subsets of the training set, enhancing robustness.
- **Model Evaluation** :
  - Performance was assessed using standard regression metrics :
    1. **Mean Absolute Error (MAE)** : Represents the average absolute difference between the actual and predicted scores. A lower value is better.
    2. **Mean Squared Error (MSE)** : The average of the squared differences between actual and predicted scores. This metric heavily penalizes larger errors. A lower value is better.
    3. **Coefficient of Determination ( $R^2$ )** : Represents the proportion of the variance in the target variable (total score) that is predictable from the independent variables. A value closer to 1.0 indicates a better fit.
  - Visualizations such as scatter plots (Actual vs Predicted Scores) and comparison of  $R^2$  are included for analysis.

## 4 Dataset Details

The dataset used for this project is a comprehensive Indian Premier League (IPL) cricket matches dataset containing detailed ball-by-ball information spanning multiple seasons.

- **Source** : The dataset was obtained from official IPL records and publicly available cricket matches data repositories.

- **Size** : The dataset contains nearly 9.5 million characters of data and covers thousands of matches and innings, making it highly representative for machine learning purposes.
- **Key Features** :
  - **Match metadata** : Season, Date, City, Venue, Match\_id
  - **Team Information** : Batting\_Team, Bowling\_Team, Toss\_Winner, toss\_decision, match\_winner
  - **Ball-by-ball Statistics** : Over, Ball, Batsman, Bowler, Runs\_Scored (batsman runs, extra runs, total runs), Wicket events, Extras (Wides, No balls, Leg byes, byes)
  - **Inning Phase Features** : Runs and Wickets per over, Runs/Wickets in last N overs
  - **Player Information** : Batsman's Name, Bowler's Name, Type of dismissal, Fielder
- **Preprocessing** :
  - All categorical variables, such as team and venue are label encoded
  - Ball-by-ball records are aggregated to compute match-level and inning-level statistics.
  - Missing or anomalous values are handled appropriately(excluded or imputed for rare scenarios).
- **Suitability** : The richness of context(recent overs, team, venue, player performance) and match outcome labels makes this dataset very well-suited for training and evaluating regression models for real-world cricket score prediction.

## 5 Comparative Analysis & Results

To evaluate the most effective model for predicting cricket scores, five different regression algorithms were trained and tested on the prepared dataset : Linear Regression, Lasso Regression (L\_1 Regularization), Ridge Regression (L\_2 Regularization), Elastic Net Regression and Random Forest Regression algorithm. The models were compared using standard performance metrics and visual analysis.

### 5.1 Model Performance Metrics

The performance of each model was quantified using Mean Absolute Error (MAE), Mean Squared Error (MSE), and the R-squared ( $R^2$ ) score.

The results from the validation and testing dataset are summarized in the table below:

| Model                | MAE         | MSE          | $R^2$         |
|----------------------|-------------|--------------|---------------|
| Linear Regression    | 11.93       | 256.19       | 0.7005        |
| Lasso Regression     | 19.83       | 642.59       | 0.2488        |
| Ridge Regression     | 12.00       | 257.60       | 0.6989        |
| Net Regression       | 21.25       | 740.61       | 0.1342        |
| <b>Random Forest</b> | <b>4.05</b> | <b>44.04</b> | <b>0.9485</b> |

Table 1: Result on validation set

| Model                | MAE         | MSE          | $R^2$         |
|----------------------|-------------|--------------|---------------|
| Linear Regression    | 12.03       | 255.22       | 0.6985        |
| Lasso Regression     | 19.87       | 645.60       | 0.2373        |
| Ridge Regression     | 12.09       | 257.23       | 0.6961        |
| Net Regression       | 21.22       | 737.24       | 0.1290        |
| <b>Random Forest</b> | <b>4.10</b> | <b>43.55</b> | <b>0.9486</b> |

Table 2: Result on testing set

- The table clearly shows, among all tested models, the Random Forest regressor significantly outperformed the linear and regularized regression models, achieving the lowest error rates on training dataset (MAE = 4.10, MSE = 43.55) and the highest  $R^2$  score (0.9486), indicating excellent predictive accuracy and strong capture of nonlinear relationships.

- Linear Regression and Ridge Regression delivered very similar results ( $MAE = 12$ ,  $R^2 = 0.70$ ), reflecting the effectiveness of simple linear modeling for this dataset where features show a direct linear relationship with the target. Lasso Regression and Elastic Net, which apply stronger regularization penalties, both performed worse than Ridge and Linear, with Elastic Net having the lowest  $R^2$  (0.1290), likely due to over-penalizing important predictors and shrinking critical coefficients toward zero.
- These results highlight that while linear models are interpretable and robust, ensemble techniques such as Random Forest provide substantial improvements in prediction accuracy for this task by capturing more complex interactions within the cricket match data.

## 5.2 Visual Analysis

Visualizations confirm the findings from the metrics, providing a clearer picture of each model's predictive accuracy.

### • Actual vs Predicted Plots :

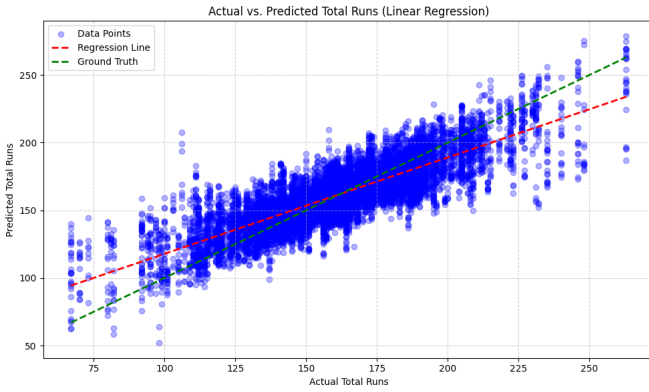


Figure 1: Linear Regression

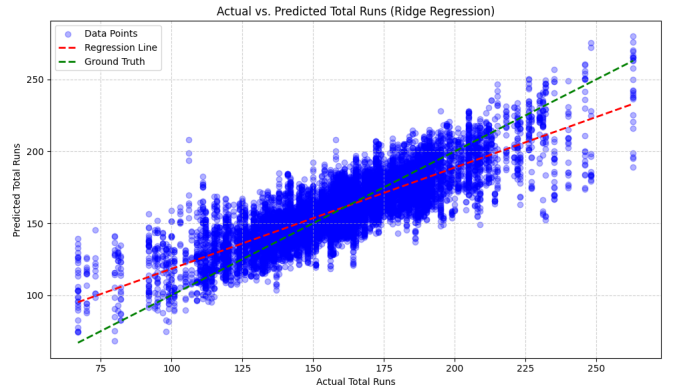


Figure 2: Ridge Regression

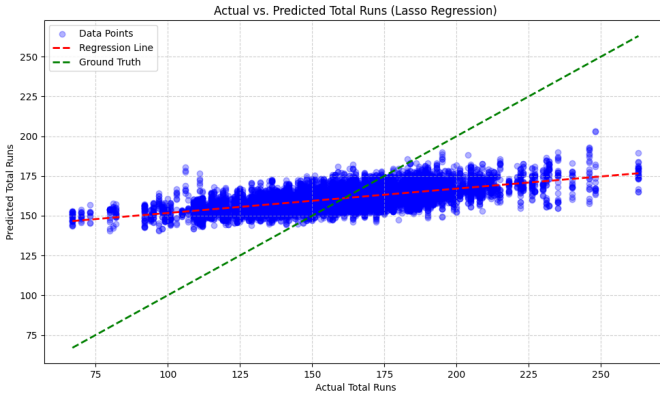


Figure 3: Lasso Regression

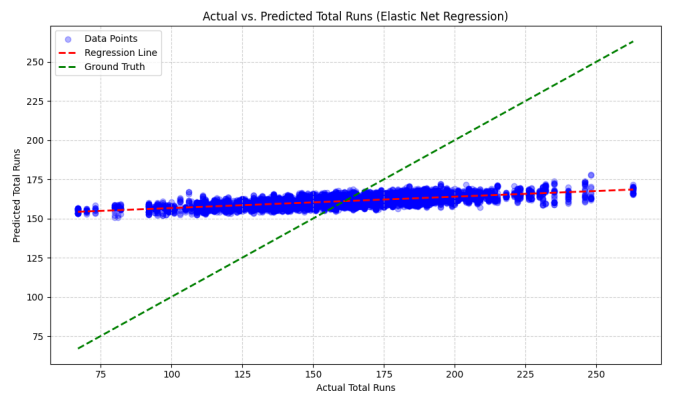


Figure 4: Elastic Net Regression

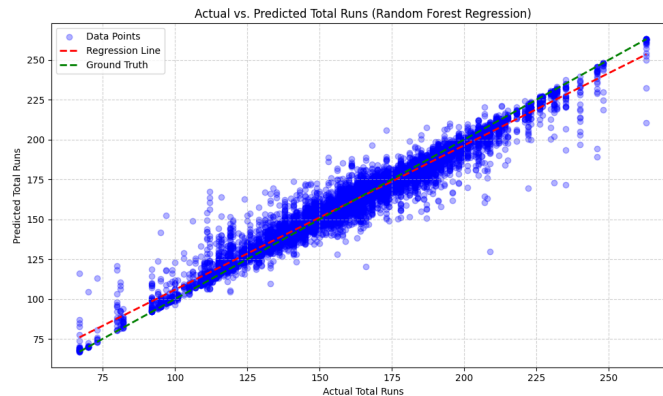


Figure 5: Random Forest Regression

These scatter plots illustrate the relationship between actual total score (x-axis) and score predicted by the model (y-axis) for each regression model. The green dotted line represents a "ground truth" where the predicted value equals the actual value.

- **Linear and Ridge Regression** : The plots for both Linear Regression and Ridge Regression yield similar patterns, with points generally distributed close to the ideal diagonal line but with visible spread, especially for extreme values.
- **Lasso and Elastic Net Regression** : Both display greater deviation from the ideal, indicating less accurate predictions due to their stronger regularization, which overly shrinks key coefficients.
- **Random Forest Regression** : The Random Forest model's scatter plot reveals a much tighter clustering of points around the diagonal, reflecting superior predictive power and ability to capture nonlinear interactions within the cricket scoring data.

### • Model R-squared Comparison :

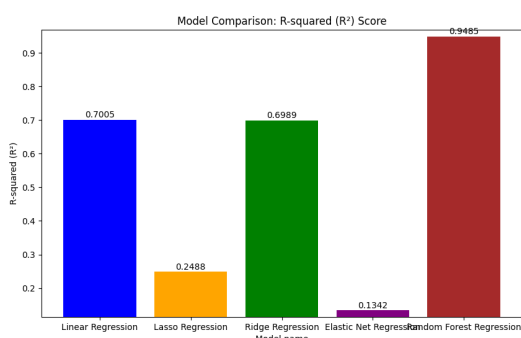


Figure 6:  $R^2$  on Validation set

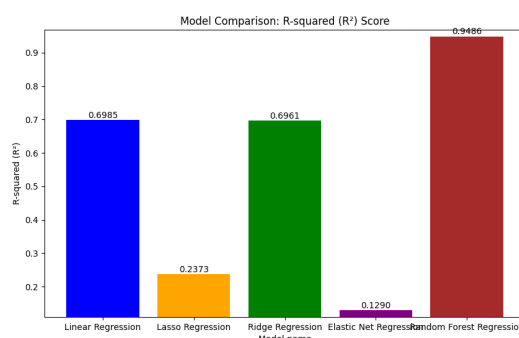


Figure 7:  $R^2$  on Testing set

This bar charts provides a direct comparison of the  $R^2$  scores. It clearly illustrates that Random Forest is the most effective predictive model among those evaluated, with nearly all observed variability in cricket score modeled accurately.

- **Feature Importance and Ablation Analysis** : To better understand which features most influence the model's predictions, a feature ablation study was conducted using the Linear Regression. The base model using all features, achieved an  $R^2$  score of 0.7005 on the validation set. Systematically dropping individual feature groups revealed the following changes in model performance :

| Dropped Feature | $R^2$  | Change in $R^2$ |
|-----------------|--------|-----------------|
| Batsman         | 0.6815 | -0.0190         |
| Bowler          | 0.6835 | -0.0170         |
| Venue           | 0.6976 | -0.0029         |
| Batting Team    | 0.6973 | -0.0032         |
| Bowling Team    | 0.6986 | -0.0019         |
| Numerical       | 0.3247 | -0.3758         |

## 5.3 Interpretation of Results

The comparative analysis demonstrates that for this specific dataset and problem, Random Forest is the superior model but Linear Regression also perform better.

- **Why Linear Regression Excelled** : The results strongly suggest that the features used (particularly runs, wickets, and overs) have a powerful, direct linear relationship with the final total score. The standard Linear Regression model was able to capture this relationship effectively without being overly complex.

- **Why Regularization Wasn't Needed :**

- **Ridge(L2) :** The Ridge model's performance was identical to Linear Regression. This implies that overfitting was not a significant issue, and the L<sub>2</sub> penalty (at an alpha of 10.0) was small enough that it didn't negatively impact the coefficients, resulting in the same predictive behavior.
- **Lasso(L1) and Elastic Net(L1+L2) :** The Lasso and Elastic Net model's poorer performance indicates that their penalty was too aggressive. By forcing some feature coefficients to shrink (potentially towards zero), it likely diminished the impact of important predictors, leading to a loss of critical information and a less accurate model.

- **Why Random Forest Excelled :** Random Forest effectively captures nonlinear patterns and complex interactions between features—capabilities that linear models lack. By constructing an ensemble of diverse decision trees, Random Forest reduces the risk of overfitting and improves predictive accuracy, especially when relationships in the data are not strictly linear. It handles both categorical and numerical features robustly and is less sensitive to multicollinearity, missing data, or outliers.

## 5.4 Error Analysis

While the models demonstrated promising predictive accuracy overall, certain prediction errors were observed under specific match conditions. For instance, matches with unusual pitch behavior or sudden weather changes led to larger discrepancies between predicted and actual scores. Additionally, innings with extraordinary individual player performances that deviated from the norm were harder to model accurately. Some error cases also highlighted limitations due to missing real-time situational factors such as runs required or wickets in hand. Analyzing these errors provides insights into improving feature selection and suggests the incorporation of live match dynamics for future model enhancements.

## 6 Conclusion

This project successfully applied and compared multiple regression techniques for cricket score prediction, ranging from basic linear approaches to advanced ensemble methods. While linear, Ridge, and Lasso regressions provided moderate accuracy, Random Forest's ensemble learning approach leverages decision trees to capture subtle dependencies and mitigate overfitting, thus enabling robust predictions even in the presence of noisy or highly interrelated features. For cricket analytics, this indicates the practical advantage of using advanced ensemble models over simple linear methods for forecasting outcomes and understanding feature contributions in real-world data. For future research and applications, integrating more diverse and granular features, considering additional ensemble or deep learning algorithms, and further optimizing feature engineering could yield even greater predictive accuracy and actionable insights for cricket analytics.

Overall, the project highlights the potential of data-driven approaches for sports analytics and offers practical insights into predicting match scores in real time.

## References

1. arindall, ipl\_data.csv (Github)
2. An Introduction to Probability Machine Learning by Kevin P. Murphy
3. scikit-learn, pandas and matplotlib documentation